

B.E. Even Semester Midterm Examination, 2024-25

Subject: Economics and Accountancy

Paper Code: HSM HU 401

Full Marks: 15 | **Time:** 1 Hour

Q1. What do you mean by engineering economics? Why engineering economics is important? [5]

Answer:

Engineering Economics is the application of economic techniques and principles to evaluate design and engineering alternatives. It bridges the gap between engineering and economics by providing a framework to quantify the costs and benefits of technical projects, ensuring that resources are allocated efficiently.

Importance of Engineering Economics:

- **Decision Making:** It helps engineers and managers choose the most financially viable option among multiple design alternatives (e.g., choosing between materials or manufacturing processes).
- **Cost Analysis:** It involves life-cycle costing, assessing not just initial capital but maintenance, operational, and salvage costs.
- **Resource Optimization:** Assists in the optimal allocation of scarce resources like capital, labor, and time to maximize profits or minimize costs.
- **Feasibility Studies:** Before a large project begins, economic principles determine if the return on investment justifies the risk and capital expenditure.

Q2. Discuss different types of market with examples. [5]

Answer:

Markets are broadly classified into four major types based on competition:

- **Perfect Competition:** A market structure with numerous buyers and sellers, homogeneous products, free entry/exit, and perfect information. Sellers are price-takers. *Example: Agricultural markets like wheat or rice.*
- **Monopoly:** A market dominated by a single seller selling a unique product with no close substitutes. There are high barriers to entry. *Example: Indian Railways or regional utility/water supply companies.*
- **Monopolistic Competition:** A market with many sellers offering differentiated products (similar but not identical). Sellers have some control over prices. *Example: Restaurants, clothing brands, or toothpaste companies.*
- **Oligopoly:** A market dominated by a small number of large firms. They are interdependent; the actions of one firm significantly affect the others. *Example: The airline industry, telecommunications (Jio, Airtel), or automobile manufacturers.*

Q3. Write short notes on GDP. [5]

Answer:

Gross Domestic Product (GDP) is the total monetary or market value of all finished goods and services produced within a country's borders in a specific time period (usually a year or a quarter).

It functions as a comprehensive scorecard of a given country's economic health.

- **Formula:**
 - **C** = Consumption (Private consumer spending)
 - **I** = Investment (Business investments in capital)
 - **G** = Government Spending (Public expenditure)
 - **X - M** = Net Exports (Exports minus Imports)
- **Types:**
 - * **Nominal GDP:** Measured at current market prices without adjusting for inflation.
 - **Real GDP:** Adjusted for inflation, providing a more accurate reflection of economic growth.

Q4. Discuss major exception of law of demand. [5]

Answer:

The Law of Demand states that price and quantity demanded are inversely related. However, exceptions occur where demand rises when the price increases:

- **Giffen Goods:** Highly inferior goods that people consume more of as the price rises. Because these goods take up a huge portion of consumer income, a price increase leaves less money for higher-quality substitutes, forcing consumers to buy more of the Giffen good (e.g., cheap staple foods during a famine).
- **Veblen Goods (Conspicuous Consumption):** Luxury goods where a higher price makes the product more desirable as a status symbol. *Example: Designer handbags, luxury cars, diamonds.*

- **Expectation of Future Price Changes:** If consumers believe a good's price will rise even further in the future, they may buy more of it now despite current price increases (e.g., gold or stocks).
- **Emergencies/Necessities:** During wars or natural disasters, consumers buy basic necessities at any price due to panic and scarcity.

Q5. Briefly discuss 'returns to scale'. [5]

Answer:

Returns to Scale is a long-run economic concept that describes how the output of a business changes when there is a proportionate change in all of its inputs (factors of production like labor and capital).

It is divided into three categories:

- **Increasing Returns to Scale:** Output increases by a greater proportion than the increase in inputs. (e.g., Doubling inputs leads to a 150% increase in output). This usually happens due to economies of scale, specialization, and better technology.
- **Constant Returns to Scale:** Output increases in exact proportion to the increase in inputs. (e.g., Doubling inputs exactly doubles the output).
- **Decreasing Returns to Scale:** Output increases by a smaller proportion than the increase in inputs. (e.g., Doubling inputs only yields a 50% increase in output). This happens due to inefficiencies, management complexities, or resource constraints (diseconomies of scale).